GENERATIVE AI WITH PYTHON

CHAPTER-4

PROMPT ENGINEERING

Book: Generative AI with Python
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Overview

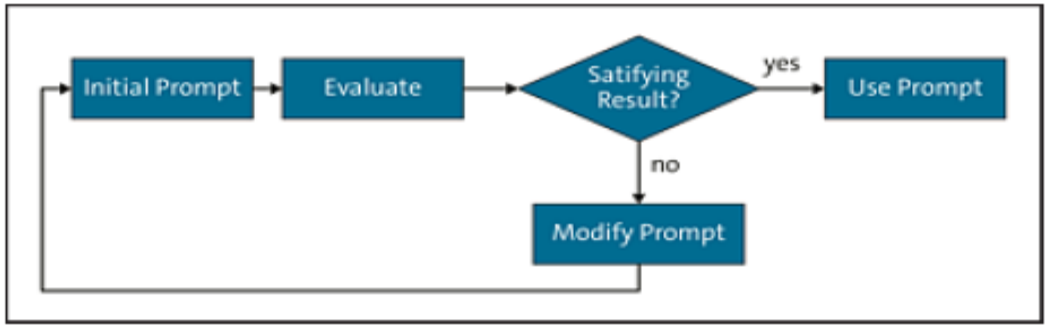
- Prompts to transform vague requests into precise instructions.
- Ultimately gives more accurate, relevant, and useful model responses.
- Represents the bridge between human intent and machine capability.
- Ability to generate effective prompts has become an important skill for developers.

Introduction

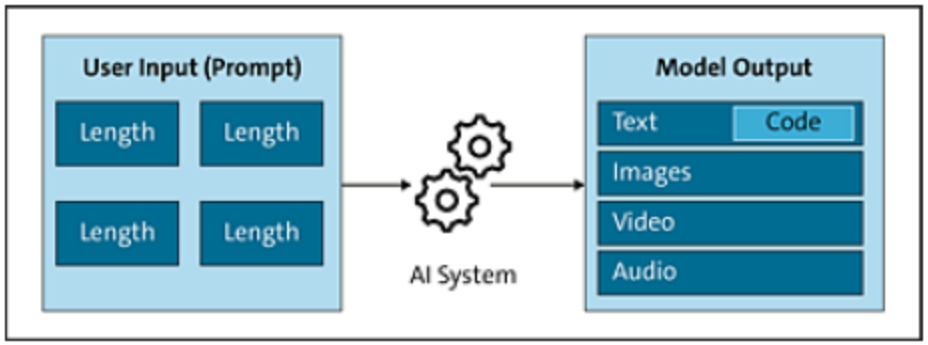
- Prompt template is converted to a prompt when invoked.



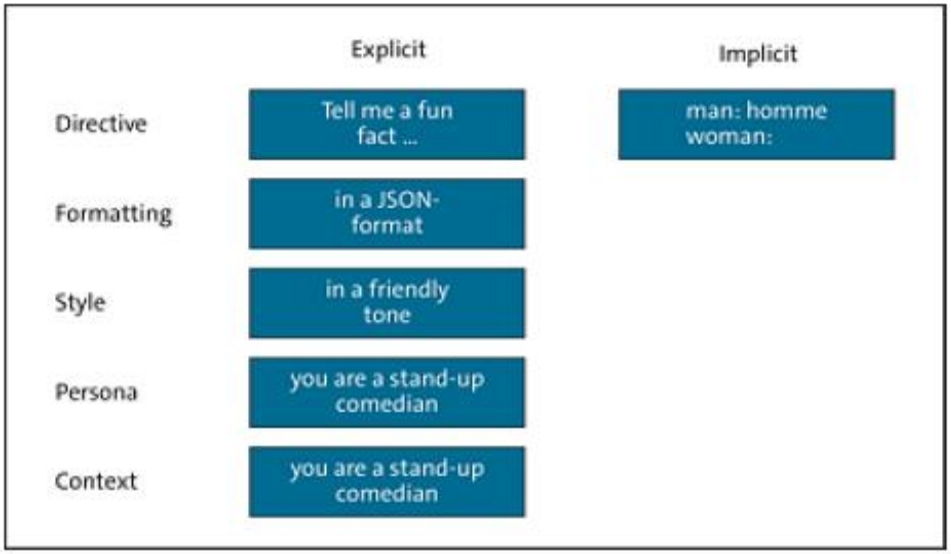
- Prompt engineering is the practice of creating inputs that guide AI systems toward a desired output.



- The AI system processes the prompt and creates an output (typically text).



- Provide pairs of words in different languages (e.g., English and Spanish)



Basic principles

1. **Clear Instructions**

Example

```
Analyze this sales data and tell me what you find.
2021: $150K
2022: $180K
2023: $165K
```

No clear metric is provided.

Improve the prompt

```
Analyze the following annual sales data from our retail store:
Sales Data:
2021: $150K
2022: $180K
2023: $165K
Please provide:
1. Year-over-year growth rates (%)
2. Identify the best and worst performing years
3. Calculate the 3-year average
4. Highlight any trends or patterns
Format: Present findings in a bullet-point list
Include: Percentage calculations and specific numbers
Context: We're a small retail business evaluating our growth trajectory
```

2. Task Decomposition

- Breaking complex tasks into smaller, manageable subtasks improves the quality and accuracy of model responses.

For example:

```
Task: Write a research paper on climate change
Steps: 1. Research the topic
2. Write the paper
3. Edit it
4. Submit
```

No, specific research focus is provided. The steps are way too general.

Improved version as:

```
Task: Write a research paper on climate change impacts on coastal cities
Phase 1: Research Planning
1.1. Topic Refinement
- Define specific research question
- Identify key variables
- Establish scope and limitations
- Create research timeline
1.2. Literature Review Preparation
- Identify key databases
- List relevant keywords
- Create citation management system
- Develop screening criteria
Phase 2: Data Collection
.......
```

3. Using Delimiters

- such as triple quotes, XML tags, or special characters help structure both inputs and outputs clearly.

4. Requesting Explanations

- Asking the model to explain its reasoning or thought. For example: including phrases like “explain your reasoning”

5. Leveraging Personas

- Assigning specific roles to the model. For example: asking the model to “respond as an experienced “write as a financial analyst”

6. Providing Examples

- like few-shot learning showing the model one or more examples of the type of output that we want.

7. Output Control

-Implementing specific parameters for output format, length, and style

8. Multiple Outputs and Selection

-Generating multiple variations of a response and selecting the best one lead to higher-quality results.

Few-shot Prompting

Customer service AI chat system

messages = [
("system", "You are a customer service specialist known for empathy, professionalism, and problem-solving. Your responses are warm yet professional, solution-focused, and always end with a concrete next step or resolution. You handle both routine inquiries and escalated issues with the same level of care.")

Instructs the model on how to behave

User

Example 1:
Customer: I received the wrong size shirt in my order #12345.

Example 2:
Customer: Your website won't let me update my payment method.

Response

I'm so sorry about the sizing mix-up with your shirt order. That must be disappointing! I can help make this right immediately. You have two options:
1. I can send you a return label and ship the correct size right away
2. I can process a full refund if you prefer

Which option works better for you? Once you let me know, I'll take care of it right away.

I understand how frustrating technical issues can be, especially when trying to update something as important as payment information. Let me help you with this step-by-step:
1. First, could you try clearing your browser cache and cookies?
2. If that doesn't work, I can help you update it directly from my end.

Could you share your account email address so I can assist you further?

New Request: {customer_request}

customer_request: "I haven't received my refund yet after returning the item 2 weeks ago."

Response:

- **I understand your concern about not receiving your refund yet. I know it can be frustrating when things don't go as expected.**
- **To look into this for you, could you please provide me with your order number and the date you returned the item?**
- **Once I have that information, I can check the status of your refund and ensure it's processed promptly.**

Chain-of-Thought

- Model struggles to find the correct answer because it jumps to conclusions rather than thinking step by step.

Question

Sofia has 7 apples in her basket. Her friend Emily gives her 3 more bags of apples. Each bag contains 4 apples. How many apples does Sofia have in total?
step 1: calculate number of apples in the 3 bags from Emily
step 2: calculate total number of apples
step 3: add results of step1 and step2

Answer 19

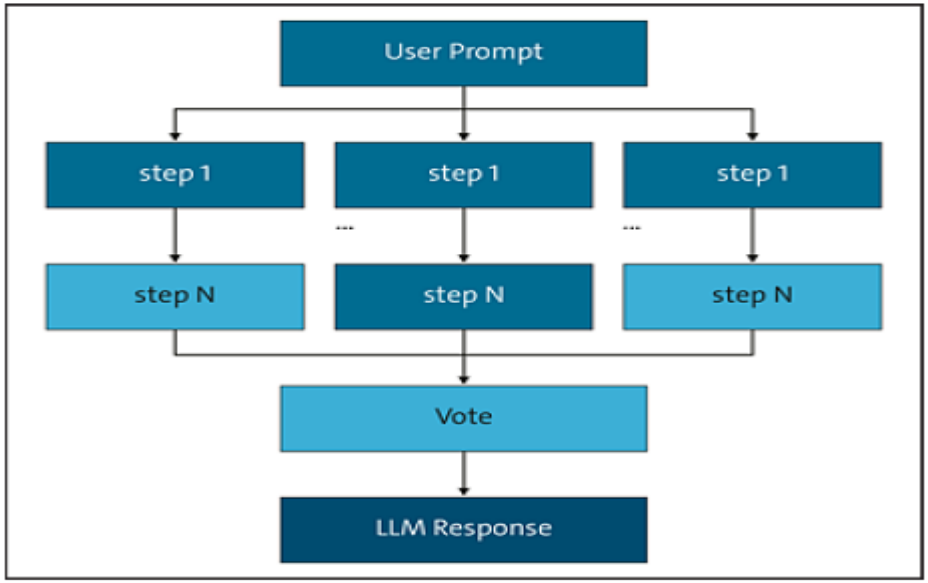
At the bakery, there are 12 cupcakes on a tray. Sarah takes 3 cupcakes for herself. Her friend Alex then takes half of the remaining cupcakes. How many cupcakes are left on the tray?

??

- Tested with model gemma-7B-it, which failed to find the right answer.

Self-Consistency Chain-of-Thought

- Running a **CoT** model several times with the same prompt
- the most popular answer is selected.
- Equivalent to *ensemble learning* in machine learning (ML).



For example: “game of 24”: 3, 4, 6, and 8

- LLMs struggle with this task.
- It is mandatory to use all four numbers.

- **Self-consistency approach** helps the AI find the right answer.
- Rather than private chain-of-thought.

  Game of 24 using 3, 4, 6, and 8 : **using Five CoT-style**

Run 1

$$(3 \times 4) \times (8 - 6)$$

$$12 \times 2 = \boxed{24}$$

Run 2

$$6 \times (3 \times 4 - 8)$$

$$6 \times (12 - 8) = 6 \times 4 = \boxed{24}$$

Run 3

$$\frac{4 + 8}{3/6}$$

$$\frac{12}{1/2} = \boxed{24}$$

Run 4

$$(6/3) \times (4 + 8)$$

$$2 \times 12 = \boxed{24}$$

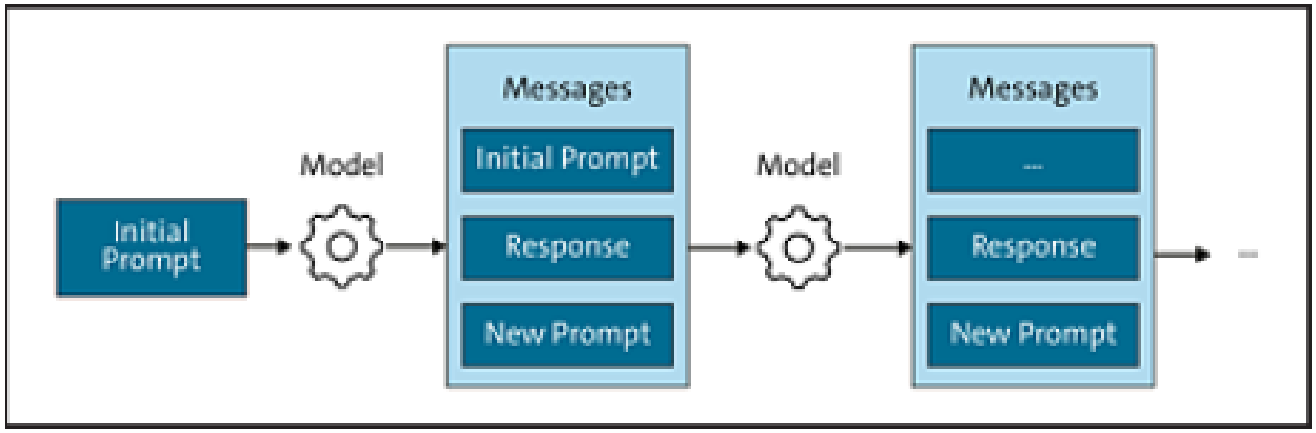
Run 5

$$4 \times (8 - 6/3)$$

$$4 \times (8 - 2) = 4 \times 6 = \boxed{24}$$

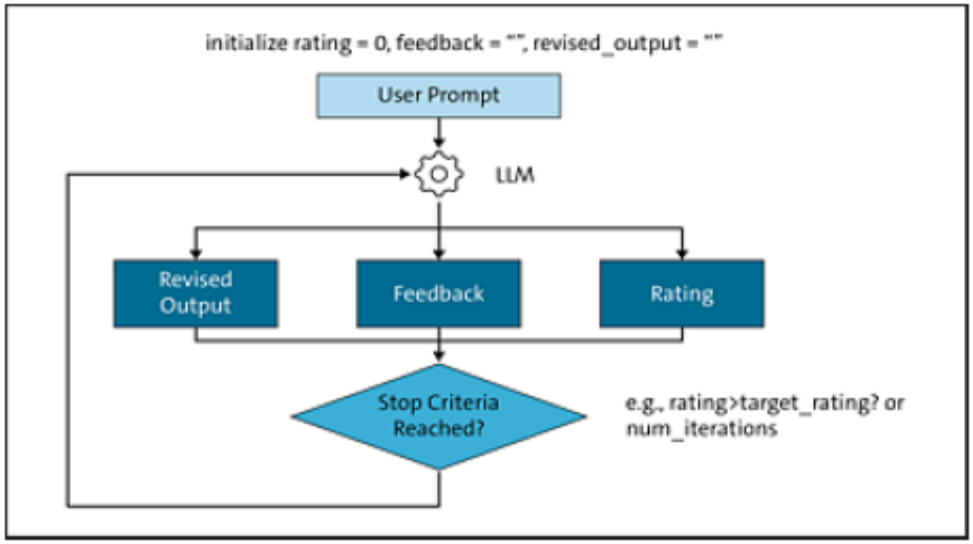
Prompt Chaining

- A technique where multiple prompts are connected in sequence to achieve complex tasks.
- Each prompt in the chain builds upon the output of the previous one.



Self-Feedback

- Empowers LLMs to evaluate responses against desired outcomes,
- Effectively enhancing the quality of their own outputs without any human intervention.
- Producing more accurate, contextually relevant, and user-aligned outputs



Thank you